

[MA323]-Applied Databases Practicum

Database-Driven Applications with GUI

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1 Preamble

This course aims to provide students with rigorous hands-on experience in the design and implementation of *database-driven applications with graphical user interfaces (GUI)*. The course integrates concepts from database systems, web technologies, and software architecture. Over the semester, students will incrementally develop a complete web-based tool, beginning with problem identification and data modeling, and culminating in advanced user interaction and the integration of both relational and non-relational databases.

The purpose of this document is to serve as a **formal blueprint for the laboratory sessions**. It specifies the project structure, expectations, evaluation methodology, and technical prerequisites in a precise and systematic manner.

2 Group Formation

- Each project group shall consist of **five (5) students**.
- Every group must nominate **one Group Leader**.
- The Group Leader will be responsible for official communication, submissions, and presentations.
- Collaboration across different groups is not permitted; however, academic discussions are encouraged.

3 Problem Identification and Scope

Each group is required to identify a **realistic and meaningful problem** that incorporates *all essential components* of this course. The proposed system must include:

1. A **relational database** with a well-designed schema (e.g., MySQL or PostgreSQL).
2. A **non-relational (NoSQL) database** (e.g., MongoDB) used in a non-trivial manner.
3. A **web-based graphical user interface**.
4. A **multi-tier architecture**, such as:

UI → Business Logic → Database Management System.

5. Explicit handling of **transactions, concurrency, and data consistency**.

Representative examples include (but are not limited to):

- Bus reservation systems with seat allocation, payment handling, and concurrent access control.
- Hotel room booking systems with authentication, availability management, and payments.
- Campus-centric applications such as hostel maintenance portals, equipment booking systems, or mess-feedback platforms.

Students are **strongly encouraged** to address *real-life problems faced within the campus*, particularly those for which a deployed solution could have tangible impact.

4 Project Registration

After forming the group, students must complete the project registration using the designated **Google Form**, available at the following link. **Only one student (the Group Leader) should submit the form on behalf of the entire group.**

<https://forms.gle/PkokPa2gLHq3PFuF8>

The submission must contain:

- Group identifier.
- Name and roll number of the Group Leader.
- Names and roll numbers of all other group members.
- A concise problem description (150–200 words) detailing:
 - The problem being solved.
 - Motivation for choosing the problem.
 - Intended users of the system.

5 Weekly Project Breakdown

Each group must prepare a tentative week-wise plan aligned with the course syllabus. A recommended reference structure is shown in Table 1.

Weeks	Expected Outcome
1–2	Finalization of problem statement, requirement analysis, and selection of technology stack.
3–4	E–R modeling, relational schema design, normalization, and manual table testing.
5–6	UI flow design, user roles definition, and MVC-based architectural planning.
7	Development of GUI forms and basic reports.
8–9	Integration of web UI with backend logic and relational database access.
10	Fully functional basic system with core features implemented.
11	UI enhancements using JavaScript and HTML5 features.
12	Asynchronous interactions using AJAX (or equivalent techniques).
13	Partial substitution or augmentation using NoSQL databases.

Table 1: Indicative week-wise project plan

6 Evaluation Scheme

The evaluation components for the course are distributed as follows:

- **Attendance:** 5%
- **Periodic Assessment:** 55%
- **End Semester Evaluation:** 40%

6.1 Periodic Assessment Structure

All groups will be partitioned into two supergroups:

- Supergroup 1: G_1, G_2, G_3, G_4
- Supergroup 2: G_5, G_6, G_7, G_8

Presentation schedule:

- Supergroup 1: Weeks 3, 5, 7, 9, and 11.
- Supergroup 2: Weeks 4, 6, 8, 10, and 12.

Each periodic assessment carries **10 points**, allocated as follows:

- Project Design: 5 points.
- Presentation Quality: 3 points.
- Technical Discussion and Question–Answer Session: 2 points.

Additional Functionality Criterion: Beyond the mandatory (basic) project requirements, groups will be awarded credit based on the *number, relevance, and complexity of extra functionalities* incorporated into their project. Projects demonstrating richer functionality, better integration, and thoughtful extensions will be evaluated more favorably within the above marking scheme.

The **final two weeks** of the semester are reserved for comprehensive project demonstrations and end-semester evaluation.

7 Technical Prerequisites

Before the subsequent laboratory sessions, students are expected to install, configure, and acquire basic working familiarity with the following software tools and technologies, which are directly aligned with the course syllabus:

- **Relational Database Management System (RDBMS):** MySQL or PostgreSQL, to support SQL queries, schema design, normalization, and transaction management.
- **NoSQL Database System:** MongoDB (or an equivalent document-oriented database), to be used during later stages of the project for partial data-model substitution or augmentation.
- **Scripting / Backend Programming Language:** PHP or Python, for implementing server-side business logic, database connectivity, and MVC-style application architecture.
- **Web Technologies:** HTML5 for structuring web pages, JavaScript for client-side logic, and AJAX for asynchronous client–server interaction.
- **Web Server Environment:** A local web server stack (e.g., Apache with PHP, or a Python-based server framework) capable of hosting and testing database-driven web applications.
- **Development Tools:** A modern web browser with developer tools enabled for debugging and performance analysis, and a source-code editor or integrated development environment (emphVisual Studio Code is recommended).

8 Concluding Remarks

Students are advised to approach the mini-project as a structured system-building exercise, analogous to a mathematical construction: clearly define entities, specify interfaces, verify prop-

erties, and iteratively refine the solution. Careful planning and disciplined execution from the first laboratory session will be critical to the success of the project.